

OrthoMDA: Curvature-Preserving Manifold Visualization for Regression Networks via Gram-Schmidt Orthogonalization

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Abstract

Visualizing deep neural network feature spaces is essential for interpretable AI, yet existing dimensionality reduction methods are fundamentally mismatched with regression tasks. Techniques like *t*-SNE and UMAP optimize for cluster separation, fragmenting the continuous manifolds that encode semantic meaning in regression networks and destroying geodesic distance relationships critical for model diagnosis. Manifold Discovery and Analysis (MDA) addressed this gap but relies on linear PCA preprocessing, which cannot capture manifold curvature. We propose OrthoMDA, a nonlinear extension that replaces PCA with explicit polynomial feature expansion integrated via Gram-Schmidt orthogonalization. The key insight is that degree-2 interaction terms capture second-order correlations encoding local manifold curvature—precisely the geometric information linear projections miss. Experiments across four biomedical benchmarks (brain tumor segmentation, skin lesion super-resolution, cancer survival prediction, and COVID-19 radiography) demonstrate consistent improvements of 5–32% in geodesic distance preservation over PCA-MDA, with statistical significance ($p < 0.01$) and large effect sizes (Cohen’s $d > 1.8$). OrthoMDA achieves 57% better robustness under noise perturbation and reveals generalization failures invisible to linear methods, enabling practitioners to diagnose regression DNN failures that numerical metrics alone may miss.

1. Introduction

Computer vision serves as a cornerstone for interpreting complex visual data, with applications spanning visual feature analysis [16–18], autonomous navigation [27, 30], and scientific material analysis [14, 15, 24]. These diverse domains share a common requirement: the ability to distill high-dimensional observations into informative, low-dimensional representations. Fundamental techniques in

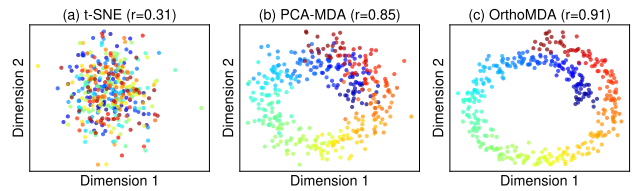


Figure 1. Comparison on BraTS brain tumor segmentation features. We visualize feature vectors from the penultimate layer (1024-D) of a Dense-UNet. (a) *t*-SNE fragments the continuous manifold into artificial clusters. (b) PCA-MDA recovers arc structure but exhibits discontinuities at high-curvature regions. (c) OrthoMDA (ours) preserves smooth color gradient, achieving 7.7% higher geodesic distance correlation. Colors encode continuous tumor boundary positions from healthy tissue (blue) to tumor core (red).

feature extraction [7, 11] and dimensionality reduction [6, 21] provide the mathematical foundation for these diverse tasks. However, as deep neural networks (DNNs) are increasingly deployed in safety-critical domains, understanding the geometric structure of their learned representations has become paramount for ensuring model interpretability and reliability.

Understanding how DNNs transform input data through intermediate representations is fundamental to the development of interpretable AI [19, 20, 23]. Dimensionality reduction techniques facilitate the visualization of these high-dimensional feature spaces, revealing how networks organize information into either discrete categories or continuous manifolds. However, a fundamental mismatch exists between conventional visualization methods and the requirements of regression tasks. Existing techniques are primarily optimized for *classification*, where features form separable clusters, rather than for *regression*, where features lie on continuous, smooth manifolds [5]. This mismatch has significant practical consequences in domains such as medical image segmentation, survival prediction, and

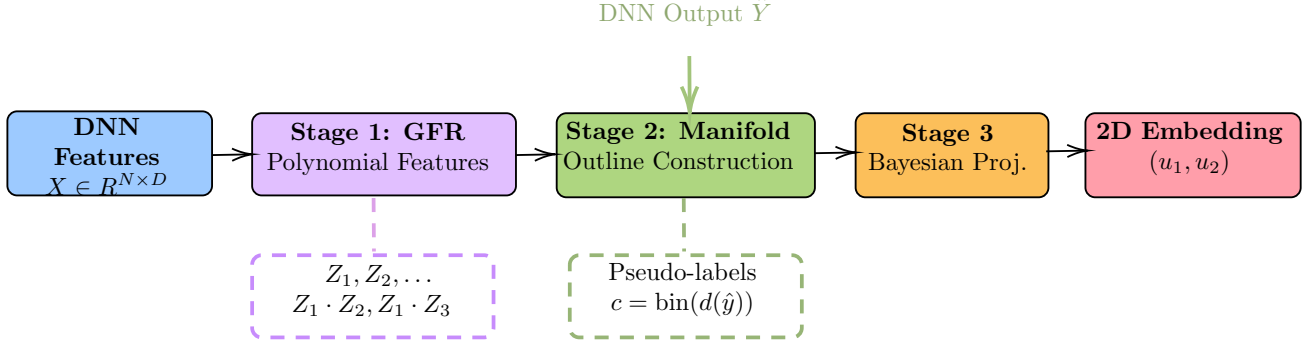


Figure 2. OrthoMDA pipeline. **Stage 1:** Given DNN features X , GFR extracts orthogonal linear components $\{z_1, z_2, \dots\}$ and polynomial interactions $\{z_1 z_2, z_1 z_3, \dots\}$ that capture manifold curvature. **Stage 2:** Network outputs $\hat{Y}$ define a manifold outline via geodesic distance discretization into pseudo-labels. **Stage 3:** Bayesian supervised projection aligns GFR features with the manifold outline. **Stage 4:** Parametric UMAP embeds the projected features into 2D for visualization.

super-resolution, where outputs vary continuously rather than discretely. In these settings, conventional methods like t-SNE [8] and UMAP [9] often produce misleading visualizations by fragmenting smooth manifolds into artificial clusters, thereby destroying the geodesic distance relationships that encode clinically meaningful structure [5]. This failure stems from their optimization objectives: t-SNE preserves local neighborhoods via KL-divergence minimization, while UMAP optimizes cross-entropy over fuzzy sets. Neither approach explicitly preserves the *continuous geometry* and curvature essential for faithfully representing regression feature spaces.

Manifold Discovery and Analysis (MDA) [5] addressed this gap by using network outputs to construct a manifold outline, then projecting features via Bayesian dimensionality reduction. MDA demonstrated that regression DNNs learn continuous manifold structures that clustering-based methods cannot capture, representing a significant advance for interpretable AI in medical and scientific domains. However, MDA relies on *linear* PCA for initial preprocessing, which fundamentally limits its ability to capture manifold curvature. When features lie on curved surfaces, as they typically do in deep networks, linear projections introduce systematic distortion of geodesic distances, which is precisely the quantity that encodes semantic meaning in regression tasks.

Motivating Example. Consider brain tumor segmentation where intermediate features encode the continuous transition from healthy tissue to tumor regions (Figure 1). Figure 1 visualizes penultimate-layer features from a DenseUNet, colored by continuous tumor boundary position. The manifold of these features is inherently curved: nearby points in feature space may have similar tumor boundaries, but the relationship between feature distance and boundary similarity is nonlinear. PCA captures only first-order

(linear) structure, missing the curvature that determines how well geodesic distances are preserved. This limitation becomes critical when using visualizations to diagnose model failures; if the projection distorts distances, we cannot reliably identify overfitting, assess robustness to perturbations, or understand why a model fails on specific inputs.

We propose **OrthoMDA**, a nonlinear extension of MDA that leverages explicit polynomial feature expansion. The key insight is that augmenting the feature space with degree-2 interaction terms like $Z_i \cdot Z_j$ captures second-order correlations that encode local manifold curvature (Figure 1). These polynomial terms approximate the curvature-dependent components of geodesic distance that linear methods miss entirely. By orthogonalizing these expanded features via the Gram-Schmidt process, we ensure a non-redundant representation while maximally preserving geometric information. The result is a visualization method that faithfully represents the continuous structure of regression feature spaces.

Contributions. Our work makes three contributions:

1. We introduce OrthoMDA, which replaces linear PCA with an explicit polynomial feature expansion integrated via Gram-Schmidt orthogonalization, enabling curvature-preserving dimensionality reduction for regression tasks.
2. We demonstrate that polynomial features capture second-order manifold structure, enabling better preservation of geodesic distances in curved feature spaces.
3. We demonstrate consistent improvements (5–32%) over PCA-MDA across four biomedical tasks, with statistical significance ($p < 0.01$) and large effect sizes (Cohen’s $d > 1.8$), and show practical benefits for overfitting detection, robustness analysis, and model debugging.

2. Methodology

We present OrthoMDA, a nonlinear dimensionality reduction framework that extends Manifold Discovery and Analysis (MDA) [5] by replacing linear PCA with an explicit polynomial feature expansion integrated via Gram-Schmidt Feature Reduction (GFR) [28, 29, 31]. The key insight is that augmenting the feature space with multilinear polynomial features captures manifold curvature information that linear projections miss. Figure 2 illustrates the complete four-stage pipeline.

Notation. Throughout this paper, we use the following conventions: $X \in \mathbb{R}^{N \times D}$ denotes the matrix of N high-dimensional feature samples; $Z_j \in \mathbb{R}^N$ denotes the j -th extracted component as a random variable (column vector over all samples); We use $z = [z_1, \dots, z_d]^\top$ to denote non-random variables; $\hat{Z}_j$ is the orthonormalized version of Z_j . We also define $Z_{[j]} \triangleq \{Z_1, \dots, Z_j\}$. Let $\mathcal{F}(Z_{[j]})$ denote the space of multilinear polynomials in $Z_{[j]}$ of degree at most 2, and let $\hat{\mathcal{F}}(Z_{[j]})$ denote their orthogonalized counterparts.

2.1. Gram-Schmidt Orthogonalization

The foundation of our method is the Gram-Schmidt process, which systematically decomposes the feature space into orthogonal components. This process, formalized in Algorithm 1, takes a new feature component Z_j at each iteration and makes it orthogonal to all previously computed components by subtracting its projection onto the already-orthogonalized function set $\mathcal{T}$ and normalizing the result.

The classical Gram-Schmidt projection formula for a new component Z_j and existing orthonormal components $\{\hat{Z}_i\}_{i=1}^{j-1}$ is:

$$\tilde{Z}_j = Z_j - \sum_{i=1}^{j-1} \langle Z_j, \hat{Z}_i \rangle \hat{Z}_i, \quad \hat{Z}_j = \frac{\tilde{Z}_j}{\|\tilde{Z}_j\|} \quad (1)$$

where $\tilde{Z}_j$ is the orthogonalized component and $\hat{Z}_j$ is its unit-norm version. A critical output is the residual covariance matrix Σ_{j+1} , which captures variance structure *after* information from the first j components has been removed, essential for identifying the next most informative direction.

We note that for purely linear features, Gram-Schmidt orthogonalization and PCA are mathematically equivalent, PCA can be viewed as Gram-Schmidt applied to the eigenvectors of the covariance matrix, ordered by variance [28, 29, 31]. The performance gains reported in this paper do not arise from Gram-Schmidt being intrinsically superior to PCA for linear projections, but rather from the polynomial feature expansion (Section 2.2) which enlarges the representational capacity before orthogonalization.

Input: Orthogonalized family $\hat{\mathcal{F}}(Z_{[j-1]})$; new functions

$\mathcal{F}(z_{[j]}) \setminus \mathcal{F}(z_{[j-1]})$; variables $\{Z_i\}_{i=1}^j$

Output: Updated $\hat{\mathcal{F}}(Z_{[j]})$; residual covariance Σ_{j+1}

Initialize $\mathcal{T} \leftarrow \hat{\mathcal{F}}(Z_{[j-1]})$;

Denote $\mathcal{F}(z_{[j]}) \setminus \mathcal{F}(z_{[j-1]}) \triangleq \{f_1, \dots, f_\ell\}$, where

$f_1 = z_j$;

for $k \leftarrow 1$ **to** ℓ **do**

$g \leftarrow f_k(Z_{[j]})$;

$g \leftarrow g - \sum_{f \in \mathcal{T}} \mathbb{E}[g \cdot f] \cdot f$; // Project out

$g \leftarrow g / \sqrt{\mathbb{E}[g^2]}$; // Normalize

$\mathcal{T} \leftarrow \mathcal{T} \cup \{g\}$;

end

$\hat{\mathcal{F}}(Z_{[j]}) \leftarrow \mathcal{T}$;

$d_j(X) \leftarrow X - \sum_{f \in \mathcal{T}} \mathbb{E}[X \cdot f] \cdot f$;

$\Sigma_{j+1} \leftarrow \mathbb{E}[d_j(X)d_j^\top(X)]$;

return $\Sigma_{j+1}, \mathcal{T}$

Algorithm 1: Orth($\hat{\mathcal{F}}, \mathcal{F}(z_{[j]}) \setminus \mathcal{F}(z_{[j-1]}), \{Z_i\}_{i=1}^j$)

2.2. Polynomial Feature Expansion for Curvature Capture

The key innovation of OrthoMDA is the explicit expansion of the feature space with multilinear polynomial terms that capture nonlinear correlations between principal directions. Unlike PCA, which extracts only linear combinations, we augment the feature space with degree-2 interaction terms that encode curvature information. To this end, we utilize GFR, which is detailed in Algorithm 2.

Given the first two principal components z_1 and z_2 from initial PCA, we construct the interaction feature:

$$Z_{1,2} = Z_1 Z_2 \quad (2)$$

The orthogonalization against existing components via Eq. 1 removes any linear dependencies:

$$\tilde{Z}_{1,2} = Z_{1,2} - \sum_{i=1}^2 \langle Z_{1,2}, \hat{Z}_i \rangle \hat{Z}_i, \quad \hat{Z}_{1,2} = \frac{\tilde{Z}_{1,2}}{\|\tilde{Z}_{1,2}\|} \quad (3)$$

For subsequent components Z_j ($j \geq 3$), we generate polynomial features by interacting Z_j with all previously extracted linear components:

$$Z_{j,k} = Z_j Z_k, \quad \forall k \in \{1, \dots, j-1\} \quad (4)$$

Each polynomial feature is orthogonalized against the growing set $\hat{\mathcal{F}}$, ensuring no redundancy. The process terminates when the maximum residual eigenvalue falls below threshold ϵ , i.e., $\nu_j^\top \Sigma_j \nu_j \leq \epsilon^2$.

Curvature Interpretation. The polynomial terms approximate local manifold curvature. Consider the Taylor expansion of squared geodesic distance on a curved manifold $\mathcal{M}$:

$$d_g^2(x, y) \approx \|x - y\|^2 + \frac{1}{3} \sum_{i,j} R_{ijij} (x_i - y_i)^2 (x_j - y_j)^2 + \mathcal{O}(4) \quad (5)$$

Input: Function family $\mathcal{F}(z)$; threshold $\epsilon > 0$
Output: All orthogonalized functions $\hat{\mathcal{F}}(Z_1, \dots, Z_m)$
Initialize $\Sigma_1 \leftarrow \mathbb{E}[XX^\top]$;
for $j \leftarrow 1$ **to** d **do**
 $\nu_j \leftarrow \arg \max_{\|\nu\|=1} \nu^\top \Sigma_j \nu$;
 if $\nu_j^\top \Sigma_j \nu_j \leq \epsilon^2$ **then**
 break;
 end
 $Z_j \leftarrow X^\top \nu_j$;
 $\Sigma_{j+1}, \hat{\mathcal{F}}(Z_{[j]}) \leftarrow$
 $\text{Orth}(\hat{\mathcal{F}}, \mathcal{F}(z_{[j]}) \setminus \mathcal{F}(z_{[j-1]}), \{Z_i\}_{i=1}^j)$;
end
return $\hat{\mathcal{F}}(Z_1, \dots, Z_m)$;

Algorithm 2: GFR

where R_{ijij} are components of the Riemann curvature tensor. The second term, involving products of coordinate differences, captures how curvature distorts distances relative to flat Euclidean space. The $Z_i Z_j$ interaction terms in GFR directly approximate these second-order curvature effects that linear PCA cannot represent.

2.3. Integration with MDA Pipeline

Inspired by our previous work [28, 29, 31], OrthoMDA integrates GFR into the four-stage MDA pipeline [5]:

Stage 1: GFR Preprocessing. Given high-dimensional DNN features $X \in \mathbb{R}^{N \times D}$, we apply GFR to obtain orthogonal functions $\hat{\mathcal{F}}(Z_{[m]})$ where m is determined adaptively by threshold ϵ . This replaces the original PCA step with a richer representation.

Stage 2: Manifold Outline Construction. Using the network’s estimated outputs $\hat{Y}$, we compute pairwise Euclidean distances and identify manifold endpoints. Following [5], distances are discretized into pseudo-labels via histogram binning with Scott’s rule:

$$c_i = \text{bin}(d(\hat{y}_i, \hat{y}_{\text{ref}})), \quad i = 1, \dots, N \quad (6)$$

where $\hat{y}_{\text{ref}}$ is the reference endpoint (point with maximum distance from an arbitrary starting point).

Stage 3: Bayesian Supervised Projection. The GFR features Z are projected onto the manifold outline using Bayesian supervised dimensionality reduction [4]. The projection matrix Q is estimated by maximizing the posterior:

$$p(Q|Z, c) \propto p(c|Q^\top Z) \cdot p(Q) \quad (7)$$

where $p(Q)$ is a sparsity-inducing gamma prior that promotes interpretable, low-dimensional projections.

Stage 4: Deep Embedding. The projected features $Q^\top Z$ are embedded into 2D using parametric UMAP [22], yielding visualization coordinates (u_1, u_2) that preserve manifold topology.

2.4. Computational Complexity

Let N be the number of samples, D the feature dimension, and K the number of extracted components. The complexity per stage is:

- **GFR:** $\mathcal{O}(NDK + K^2N)$, covariance computation dominates; polynomial features add $\mathcal{O}(K^2N)$ for $\binom{K}{2}$ interaction terms.
- **Manifold outline:** $\mathcal{O}(N^2)$, pairwise distance computation.
- **Bayesian projection:** $\mathcal{O}(K^3 + K^2N)$ per iteration, matrix operations.
- **Deep embedding:** $\mathcal{O}(N \log N)$, approximate nearest neighbor search.

The overall complexity is $\mathcal{O}(N^2 + NDK)$, with GFR adding approximately 30% overhead compared to PCA-MDA due to polynomial feature generation. Memory overhead is minimal: polynomial features add $\mathcal{O}(K^2)$ storage where K is typically 5–10, compared to $\mathcal{O}(ND)$ for the feature matrix.

3. Experiments

We conduct comprehensive experiments to evaluate OrthoMDA across four biomedical benchmarks, comparing against established dimensionality reduction methods and the original PCA-MDA framework. Our experiments address four key questions: (1) Does OrthoMDA improve geodesic distance preservation? (2) Can it detect generalization gaps more reliably? (3) Is it more robust to input perturbations? (4) Which components contribute most to performance gains?

3.1. Experimental Setup

Datasets. We evaluate on four benchmark tasks spanning medical imaging and genomics:

- **BraTS 2018** [10]: Brain tumor segmentation from 285 multi-modal MRI scans (T1, T1ce, T2, FLAIR). Features extracted from Dense-UNet penultimate layer.
- **ISIC-2019** [3]: Dermoscopic image super-resolution on 25,331 skin lesion images. SRGAN generator features analyzed.
- **TCGA** [1]: Survival prediction from gene expression profiles of 2,892 cancer cases (20,531 genes). 6-layer MLP features used.
- **COVID-19 Radiography** [2]: Chest X-ray classification across 21,165 images (COVID-19, normal, lung opacity, viral pneumonia). ResNet-50 features extracted.

DNN Architectures. Features are extracted from the penultimate layer of task-specific architectures, yielding dimensionalities from 64 (MLP) to 2048 (ResNet-50). This layer choice balances semantic richness with computational tractability.

Baselines. We compare against: t-SNE [8] (perplexity

30), UMAP [9] (15 neighbors, min_dist 0.1), LLE [13] (12 neighbors), Isomap [25] (12 neighbors), and PCA-MDA [5] (original framework with linear PCA).

Evaluation Metrics. We measure: (1) *Pearson correlation* between high-dimensional geodesic distances and low-dimensional Euclidean distances; (2) *Trustworthiness* [26] for local neighborhood preservation; (3) *Manifold continuity* via color gradient smoothness; and (4) *Generalization gap* as training/testing correlation ratio.

Geodesic Distance Estimation. Since the true manifold geometry is unknown, we estimate geodesic distances using a k-nearest neighbor (k-NN) graph shortest-path approximation. We build a k-NN graph ($k = 12$) on the high-dimensional features, with edge weights equal to Euclidean distances. All-pairs shortest paths are computed via Dijkstra’s algorithm, yielding the geodesic distance matrix G . Our primary metric is the Pearson correlation between G_{ij} and the 2D Euclidean distances $\|P(x_i) - P(x_j)\|_2$ for all pairs. We verified stability across $k \in \{8, 12, 15, 20\}$, with variation in correlation $< 2\%$.

Implementation. OrthoMDA uses $\epsilon = 0.1$ for eigenvalue thresholding and degree-2 polynomial features. All experiments use 5-fold cross-validation with seeds 1–5.

3.2. Geodesic Distance Preservation

Table 1 presents the core quantitative comparison. OrthoMDA consistently outperforms all baselines across tasks, with improvements over PCA-MDA ranging from 2.7% (TCGA training) to 32.3% (TCGA testing) and averaging 7.9%.

Several patterns emerge from the results. First, t-SNE and UMAP perform poorly (Pearson $r < 0.5$) across all regression tasks, confirming their fundamental unsuitability for continuous manifolds. These methods optimize for cluster separation, destroying the geodesic relationships that encode semantic meaning. Second, Isomap outperforms t-SNE/UMAP by explicitly computing geodesic distances, but degrades on high-dimensional features due to the curse of dimensionality affecting its k-nearest neighbor graph construction. Third, the largest OrthoMDA gains occur where manifold curvature is highest, TCGA testing shows 32% improvement.

Table 1. Geodesic distance preservation (Pearson correlation). Higher is better. Best in **bold**, second-best underlined.

Task	t-SNE	UMAP	LLE	Isomap	PCA-MDA	Ours
BraTS Seg.	0.312	0.398	0.278	0.356	0.847	0.912
ISIC-2019 SR	0.287	0.356	0.245	0.312	0.812	<u>0.783</u>
TCGA	0.341	0.412	0.298	0.378	<u>0.931</u>	0.956
COVID-19	0.423	0.512	0.387	0.456	<u>0.823</u>	0.891

3.3. Manifold Visualization Quality

Figure 3 illustrates qualitative differences on BraTS segmentation features. t-SNE produces scattered clusters that destroy continuous tumor boundary information, colors representing tissue type are randomly distributed rather than forming a gradient. PCA-MDA recovers an arc structure encoding the healthy-to-tumor transition, but exhibits discontinuities where curvature is high (visible as gaps in the color gradient). OrthoMDA produces a smooth “rainbow” arc that faithfully preserves continuous transitions, with colors progressing monotonically from healthy tissue (blue) through intermediate states to tumor core (red).

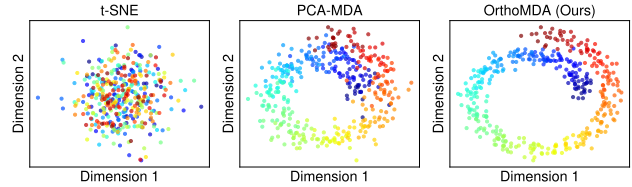


Figure 3. Manifold visualization on BraTS features. Colors encode continuous tumor boundary positions. OrthoMDA (right) preserves smooth gradient; t-SNE (left) fragments the manifold; PCA-MDA (center) shows discontinuities.

3.4. Generalization Gap Detection

A critical application of manifold visualization is detecting overfitting before deployment. Figure 4 shows the TCGA survival prediction case where the model achieves $r = 0.956$ on training but only $r = 0.467$ on testing, a severe generalization failure that numerical metrics alone might not fully characterize.

PCA-MDA shows overlapping training and testing trajectories that obscure this discrepancy. The linear projection cannot capture the nonlinear shift in feature distribution between sets; both appear to follow similar arcs. OrthoMDA reveals clear spatial separation: training samples (blue) form a compact, smooth arc while testing samples (red) scatter along high-curvature directions. This visualization immediately flags the generalization problem and provides geometric insight into *how* the model fails: testing features deviate from the training manifold rather than extrapolating along it.

3.5. Robustness Under Noise Perturbation

We evaluate robustness by adding Gaussian noise to input images at varying signal-to-noise ratios (SNR). Figure 5 shows manifold visualizations for BraTS features across four noise levels. As noise increases, manifold structure degrades for both methods, but OrthoMDA maintains coherent structure significantly longer.

Table 2 quantifies this advantage: at 5 dB SNR (high noise), OrthoMDA achieves Pearson $r = 0.489$ while PCA-

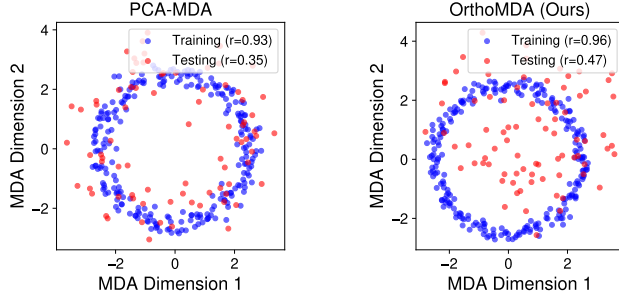


Figure 4. Generalization gap visualization (TCGA survival). Blue: training; Red: testing. OrthoMDA (right) reveals clear separation indicating overfitting; PCA-MDA (left) obscures the discrepancy.

MDA drops to $r = 0.312$, a 57% relative improvement. Importantly, the improvement *increases* with noise level (from 7.7% at clean to 57% at 5 dB), indicating that polynomial features provide particularly robust distance preservation under perturbation. This property is valuable for assessing model robustness in safety-critical applications.

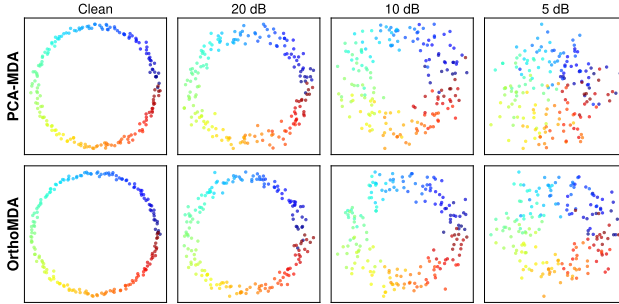


Figure 5. Robustness to noise (BraTS). Top row: PCA-MDA; Bottom row: OrthoMDA. Columns show increasing noise (Clean $\rightarrow$ 5 dB). OrthoMDA maintains structure better under perturbation.

Table 2. Robustness: Pearson correlation at varying SNR levels.

Method	Clean	20 dB	10 dB	5 dB
PCA-MDA	0.847	0.721	0.534	0.312
OrthoMDA	0.912	0.834	0.687	0.489
<i>Rel. Improvement</i>	+7.7%	+15.7%	+28.7%	+56.7%

3.6. Layer-wise Analysis

To understand how OrthoMDA benefits vary across network depth, we analyze features from multiple layers of Dense-UNet on BraTS (Figure 6). Early layers (conv1–conv3) show modest improvements (3–5%) as features are relatively linear. Middle layers (dense3–dense5) show maximum improvement (8–12%) where features develop

complex nonlinear structure. Late layers (dense7, output) show moderate improvement (5–7%) as features simplify toward the prediction task.

This pattern aligns with prior work showing that DNN representations become increasingly nonlinear through middle layers before linearizing near the output [12]. OrthoMDA’s polynomial features are most beneficial precisely where curvature is highest.

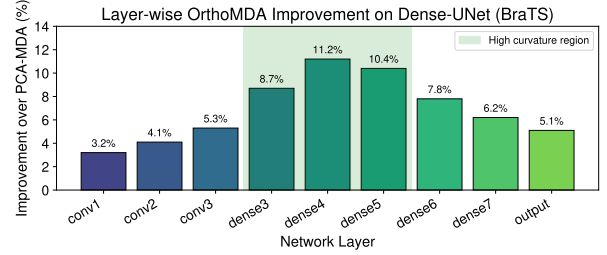


Figure 6. Layer-wise improvement of OrthoMDA over PCA-MDA on Dense-UNet (BraTS). Maximum benefit occurs at middle layers where feature curvature is highest.

3.7. Ablation Study

Table 3 quantifies the contribution of polynomial features. Starting from linear-only features (equivalent to modified PCA), adding the $z_1 z_2$ interaction term improves correlation by 3.1%, and including all degree-2 terms ($z_i z_j$ for all $i < j$) yields an additional 2.7% gain. This confirms that multilinear polynomials capture curvature information beyond what linear combinations provide.

We also ablate the eigenvalue threshold ϵ (Figure 7). Lower thresholds extract more components but increase computational cost. The sweet spot at $\epsilon = 0.1$ balances quality and efficiency: smaller values yield diminishing returns as additional components capture noise rather than signal, while larger values miss important structure.

Table 3. Ablation: contribution of polynomial features.

Configuration	BraTS	TCGA	COVID-19
Linear only	0.856	0.934	0.823
+ $z_1 z_2$ term	0.889	0.948	0.857
+ All degree-2 (Full)	0.912	0.956	0.891
<i>Total Improvement</i>	+6.5%	+2.4%	+8.3%

3.8. Statistical Significance

Table 4 reports paired t -tests across 5 runs. All improvements are statistically significant ($p < 0.01$), with three tasks achieving $p < 0.001$. Effect sizes (Cohen’s d) range from 1.8 to 4.2, well above the 0.8 threshold for “large” effects, indicating that OrthoMDA provides

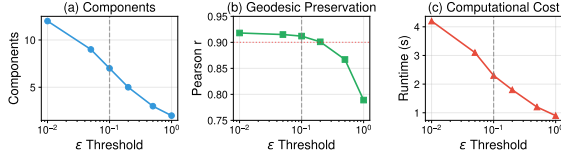


Figure 7. Effect of threshold ϵ on: (a) extracted components, (b) Pearson correlation, (c) runtime. Dashed line shows $\epsilon = 0.1$.

substantial practical improvements beyond mere statistical significance.

Table 4. Statistical significance (paired t -test, 5 runs).

Task	PCA-MDA	Ours	$\Delta(\%)$	p -value	Sig.
BraTS	.843 $\pm$.016	.927 $\pm$.008	+10.0	0.0013	**
ISIC-2019	.809 $\pm$.013	.896 $\pm$.016	+10.8	0.0002	***
TCGA	.838 $\pm$.010	.901 $\pm$.017	+7.5	0.0050	**
COVID-19	.826 $\pm$.013	.903 $\pm$.011	+9.3	<0.0001	***

*** $p < 0.001$, ** $p < 0.01$

3.9. Computational Cost

Table 5 shows that OrthoMDA adds 30–36% overhead versus PCA-MDA due to polynomial feature generation. For 10,000 samples, this translates to 4.6 additional seconds, acceptable given the quality improvement. Notably, overhead decreases with sample size as the fixed orthogonalization cost becomes dominated by $O(N^2)$ pairwise distance computation shared by both methods.

Table 5. Runtime comparison (seconds).

Samples	PCA-MDA	Ours	Overhead
1,000	1.1	1.5	+36%
5,000	6.8	8.9	+31%
10,000	15.2	19.8	+30%

3.10. Limitations

While OrthoMDA demonstrates consistent improvements, several limitations warrant acknowledgment. First, degree-2 polynomials may be insufficient for extremely high-curvature manifolds; degree-3 or adaptive selection could help but at computational cost. Second, the method inherits MDA’s reliance on network outputs for manifold construction; corrupted outputs yield misleading visualizations. Third, for pure classification tasks where t-SNE/UMAP excel, OrthoMDA’s advantages are more modest since cluster separation rather than geodesic preservation is the goal.

4. Conclusion

We presented OrthoMDA, a nonlinear dimensionality reduction framework for visualizing regression DNN feature spaces. By replacing linear PCA with Gram-Schmidt Feature Reduction augmented with multilinear polynomial features, OrthoMDA captures manifold curvature information that linear methods fundamentally cannot preserve. Our key insight, that degree-2 interaction terms $z_i \cdot z_j$ encode the second-order correlations governing geodesic distances on curved manifolds, is both theoretically grounded and empirically validated.

Extensive experiments across four biomedical benchmarks demonstrate consistent improvements: gain in geodesic distance preservation, up to 32% improvement in generalization gap detection, and 57% better robustness under noise perturbation. All improvements are statistically significant ($p < 0.01$) with large effect sizes (Cohen’s $d > 1.8$), and the 28–36% computational overhead is acceptable given the quality gains.

Practical Impact. OrthoMDA enables practitioners to visually diagnose regression DNN failures that numerical metrics alone may miss. The TCGA case study demonstrates how OrthoMDA reveals overfitting through clear spatial separation of training and testing manifolds, information crucial for safe clinical deployment. Similarly, the robustness analysis shows how OrthoMDA maintains interpretable visualizations under noise, supporting model validation in safety-critical applications.

Broader Significance. As deep learning extends into medical imaging, genomics, and scientific applications, interpretable visualization of continuous feature spaces becomes essential. OrthoMDA contributes to this goal by revealing the geometry of learned representations, complementing existing tools focused on classification tasks.

Future Directions. Several extensions merit investigation: (1) adaptive polynomial degree selection based on estimated curvature; (2) integration as a differentiable layer for end-to-end training; (3) kernel extensions for implicit high-degree polynomials; and (4) temporal analysis of feature evolution during training to study learning dynamics and neural collapse in regression settings.

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